

Dexter Fernandes

Bristol • +44 07436007369 • dexterfernandes11@gmail.com • linkedin.com/in/dexterferns •
<https://github.com/Dexter-Fernandes>

Computer Vision Engineer

Computer Vision Engineer with 3+ years of experience building and shipping production perception systems across the full lifecycle, from dataset curation and training pipelines through deployment, validation, and model performance monitoring. Experienced in benchmarking current computer vision models across detection, tracking, segmentation, and depth estimation, with a practical understanding of accuracy, latency, and deployment footprint trade-offs. MSc Robotics graduate from the University of Bristol, with dissertation work in ROS2, LiDAR SLAM, point cloud processing, and sensor synchronisation.

Stack: Python, C++17, PyTorch, OpenCV, OpenVINO, MLflow, DVC, Airflow, Docker.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Owned the end-to-end computer vision pipeline for real-time perception systems in harsh industrial environments, from data curation and model training through deployment and on-site validation, improving F1 by 15% while maintaining deterministic 24/7 operation.
- Orchestrated a sensor-data cleansing pipeline for 3.6 million records using Pandas and Apache Airflow, reducing model rollout time from 3 days to under 24 hours.
- Built multimodal data ingestion, cleansing, training, and validation pipelines across image, video, and sensor data using Python, Pandas, Airflow, MLflow, DVC, Optuna, and CVAT, cutting model iteration time by 40% and reducing rollout time from 3 days to under 24 hours.
- Refactored legacy services into Dockerised microservices with Kafka-based messaging, structured logging, health checks, retries, and safe restarts, improving fault isolation, increasing MTBF by 35%, and reducing live video pipeline crashes by 67%.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, improving maintainability and reducing defects.
- Implemented Git and Jenkins-based continuous integration and validation workflows for edge inference modules for 24/7 deployments to improve release consistency across hardware targets.
- Collaborated with cross-functional engineering and client teams to translate performance metrics into actionable deployment and optimisation decisions.
- Validated deterministic model performance under extreme field conditions at a British Petroleum site, ensuring reliability across 24/7 operations.
- Built synthetic image generation and augmentation procedures with OpenCV and scikit-image to improve model robustness under harsh industrial conditions, reducing false positives and supporting long-term operational reliability.
- Presented key findings and metrics to leadership and clients to translate complex performance results into actionable insights for deployment and further optimisation.
- Diagnosed and resolved production failures through on-site analysis and structured validation checklists, reducing crash frequency by 67% in live video pipelines.
- Built a Django-based internal computer vision application with a web dashboard to visualise inference, system health, and performance metrics, enabling faster debugging.
- Designed adversarial training experiments (occlusion, lighting variation, motion artefacts), benchmarking robustness via F1 and failure-rate metrics, achieving a 15% F1-score improvement within 3 months.
- Developed a repeatable optimisation workflow (baseline to FP16 to validation) with defined acceptance thresholds, enabling consistent latency, memory, and stability outcomes across deployments.

- Built benchmarking and profiling harnesses to evaluate open-source models (YOLOv5, TensorFlow Model Zoo) and quantify accuracy, latency, and stability pre and post optimisation for regression detection.
- Diagnosed and resolved production issues in Dockerised microservices, improving service continuity through stronger logging, health checks, safe restarts, and workload isolation.
- Applied image preprocessing and artefact-detection workflows with OpenCV and scikit-image to identify mud and sand contamination on camera feeds, reducing false positives by 13% in harsh industrial environments.
- Applied statistical and experimental methods including Bayesian optimisation, ablation studies, hypothesis testing, and confidence intervals to support model selection, threshold tuning, and failure analysis in production.
- Worked with cross-functional engineering and client teams to translate performance findings into deployment and optimisation decisions for production systems.
- Debugged training dynamics and convergence issues in production CV models, balancing model complexity, augmentation, and validation strategy to improve generalisation under lighting variation, occlusion, motion artefacts, and camera contamination.

LivNSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a team to design, train, and deploy production object detection and tracking models for industrial safety applications, reducing false positives by 33% while meeting latency and power constraints.
- Built low-latency perception pipelines using OpenVINO, and C++17, tripling throughput from 5 FPS to 15 FPS and delivering deterministic real-time inference on Intel VPU hardware under tight power and memory constraints.
- Deployed quantised vision models to rugged Intel VPU edge devices using OpenVINO, delivering deterministic real-time inference across RGB and IR sensor streams in harsh industrial environments.
- Built automated labelling, dataset versioning, and performance-tracking workflows, repurposing existing detection models for annotation and cutting manual labelling effort by 80% while accelerating iteration cycles on large real-world datasets.
- Developed synthetic data generation and augmentation tools to improve edge-case coverage and cut manual labelling time by 60% using OpenCV and Albumentations.
- Conducted systematic quantisation and optimisation experiments benchmarking accuracy–latency–memory trade-offs and reducing model size by 50% while meeting real-time constraints.
- Developed adversarial-condition training and validation conditions simulating occlusion, lighting variation, and motion artefacts, cutting performance variance by 25% and increasing resilience.
- Collaborated with hardware, software, and QA teams to integrate CV models into deployed systems, and presented performance findings to leadership and clients to support rollout, optimisation, and reliability decisions.
- Engineered production-grade edge perception software for safety-critical environments, delivering deterministic real-time behaviour on Intel VPU devices.
- Mentored junior engineers and collaborated with hardware, software, and QA teams to integrate perception modules into deployed systems and improve delivery consistency.
- Presented key findings and metrics to leadership and clients, translating complex performance results into actionable insights for deployment and further optimisation.
- Built and maintained large-scale image and sensor datasets, automating labelling, versioning, and performance tracking to support anomaly investigation and continuous improvement cycles.
- Defined and standardised FP16/INT8 optimisation paths (including QAT, pruning, and layer fusion), reducing model size by 50% while preserving accuracy under tight memory and latency constraints.
- Built asynchronous inference pipelines using OpenVINO and C++17 to maximise throughput on Intel VPU edge hardware under real-time latency and power constraints.
- Evaluated edge versus cloud inference trade-offs to minimise bandwidth, latency, and operational cost while maintaining safety-critical performance, supporting production cloud-hosted ML workloads on Microsoft Azure.
- Built a low-latency, high-reliability computer vision PoC for a zinc ore refining plant in South Africa, and developed a user-facing hazard detection application for an Oman-based client.
- Built and maintained a real-time multi-camera perception pipeline on a moving construction vehicle, synchronising camera streams and running CV-based incident detection across both feeds under edge compute constraints.

- Explored semantic and instance segmentation models, including DeepLabv3+, Mask R-CNN, U-Net-style architectures, and SegFormer, to support annotation review, foreground/background separation, and failure analysis for industrial camera feeds.
- Built YOLOv5-based visual quality inspection for John Deere engine paintjobs on a conveyor line, detecting paint defects, contamination, and colour inconsistency as part of a manufacturing anomaly detection workflow.
- Used confusion-matrix-driven failure clustering to diagnose recurring CV failure modes, prioritise dataset fixes, and improve model robustness across camera feeds.

PROJECTS

Modern Edge CV Benchmarking

Mar 2026

- Benchmarked modern computer vision models including YOLOv8, YOLOv10, RT-DETR, EfficientViT, Grounding DINO, MobileSAM, and Depth Anything v2 to compare accuracy, latency, model size, and deployment suitability.
- Evaluated Vision Transformer and foundation-model approaches for object detection, segmentation, depth estimation, and open-vocabulary visual understanding, with a focus on production-style CV trade-offs.
- Exported models to ONNX and evaluated OpenVINO deployment paths, strengthening practical understanding of model optimisation, inference acceleration, and edge deployment constraints.
- Built benchmark comparisons using mAP, inference time, and model-size measurements to support model selection for low-latency visual AI systems.

ROS2 SLAM & Nav2 Simulation

Feb 2026

- Built a simulation-based robotics stack in ROS2 Jazzy and Gazebo Harmonic to evaluate LiDAR-centric SLAM, LiDAR+RGB fusion SLAM, and autonomous navigation in structured scenarios.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer, and GLIM across 2D and 3D LiDAR configurations, comparing mapping and localisation behaviour to support system selection and iteration.
- Developed ROS2 nodes, launch files, and parameterised configurations in C++ and Python, using tf2, message synchronisation, camera and sensor ingestion, and timestamp alignment to maintain frame consistency and debug transform timing issues.
- Implemented point cloud processing with PCL and Open3D for filtering, segmentation, and registration, and added an EKF-based localisation layer with robot_localization to fuse wheel odometry and IMU for smoother odom-to-base tracking during LiDAR-only SLAM.
- Containerised the stack with Docker on Linux and managed versioning in Git to support reproducible builds and consistent execution across development environments.

End-to-End Damage Severity & Claim Assessment

Nov 2025

- Developed an automotive claims assessment pipeline combining computer vision-based damage detection with severity classification to support faster triage of real-world vehicle damage scenarios.
- Built a multimodal image-to-structured-output workflow that converted vehicle-damage detections and severity predictions into insurer-friendly claim summaries using structured reasoning and LLM-generated explanations.
- Designed the full workflow across data preprocessing, model training, evaluation, and a user-facing inference module, demonstrating end-to-end delivery from model output to usable product insight.
- Created interpretable repair insights from CV model outputs, translating visual predictions into clear claim explanations for non-technical users.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Radiological SLAM with LiDAR Odometry. Designed and implemented a real-time SLAM system using RTAB-Map on the Clearpath Husky platform, integrating LiDAR and radiological data for localisation and mapping in inspection scenarios.
- Evaluated RTAB-Map, LIO-SAM, and Cartographer across comparable scenarios using ROS2 and rosbag2-based workflows to assess mapping and localisation behaviour.
- Validated sensor placement for coverage and calibrated LiDAR extrinsics to the robot base, including the Velodyne VLP-16 and Intel RealSense D435i.
- Implemented a ROS2 (rclcpp) point cloud pipeline in C++17 using PCL, and developed ROS2 nodes with tf2, message synchronisation, and sensor data playback workflows to maintain frame consistency and debug transform timing issues.
- Worked with camera and sensor ingestion in ROS2 pipelines, including timestamp alignment and downstream processing for SLAM and perception tasks.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.

CERTIFICATIONS

Building RAG Agents with LLMs

Nvidia

Mar 2026 - Apr 2026

Deep Learning Specialisation

deeplearning.ai

Jul 2019

SKILLS

- **SLAM, Camera and Data Pipelines:** Camera ingestion, Cartographer, Ceres, Eigen, Filtering, GLIM, LiDAR SLAM, LIO-SAM, Multi-camera perception, Point cloud processing, Pose graph optimisation, Registration, RTAB-Map, Scan matching, Segmentation, Video and sensor data processing pipelines, Visual data streams
- **Cloud, Edge & MLOps:** Apache Airflow, Apache Kafka, API design, AWS, CI/CD, Docker, DVC (Data Version Control), Health checks, Intel VPUs, Microsoft Azure, MLFlow, Post-deployment validation, Production monitoring, Structured logging
- **Programming & Tooling:** Bash, C++17, CMake, colcon, Django, FastAPI, Git, Intel VTune, Linux, NumPy, Pandas, PG vector, PostgreSQL, Python, Qt, Scikit-Learn
- **Deep Learning and Optimisation:** Bayesian Optimisation, Convolutional Neural Networks, Data Augmentation, FP16/INT8 deployment, Layer fusion, Object Detection & Tracking, ONNX, OpenVINO, Optuna, Post-training Quantisation, PyTorch, Quantisation-Aware Training, Scikit-Learn, Structured pruning, TensorFlow, TensorRT, YOLO
- **Edge AI and Inference:** Intel Core i7 (VTC7252-7C4IP), Intel Movidius Myriad VPU, NVIDIA Jetson AGX Xavier
- **Professional Skills:** Agile framework, Experiment-driven development, Stakeholder communication, Technical leadership
- **Point Cloud & Vision Processing:** Denoising, Open3D, OpenCV, Point Cloud Library, Scikit-Image
- **Robotics and Middleware:** Closed-loop robot control, Gazebo Harmonic, Inverse kinematics, launch files, message synchronisation, Nav2, rclcpp, rclpy, Real-time perception systems, robot_localization, ROS2 (Robot Operating System 2), rosbag2, sensor data playback, Sensor integration, tf2, timestamp synchronisation

- **Computer Vision & Image Processing:** Anomaly detection, Data augmentation, Defect detection, Image processing, LiDAR extrinsics calibration, Object detection, Object tracking, Open3D, OpenCV, PCL (Point Cloud Library), PIL, scikit-image, Segmentation, Sensor calibration, Visual quality inspection
- **AI-Assisted Development:** ChatGPT Codex, Claude Code, Prompt engineering
- **Computer Vision Models:** ByteTrack, DeepLabv3+, Depth Anything v2, EfficientDet, EfficientViT, Faster R-CNN, Grounding DINO, Mask R-CNN, MobileSAM, RT-DETR, SegFormer, SSD MobileNet V2, U-Net, YOLOv10, YOLOv5, YOLOv8
- **Datasets, Annotation & Evaluation:** Automated labelling, Cityscapes, COCO, Confusion-matrix-driven failure clustering, CVAT, DVC, F1-score, ImageNet, IoU, IoU, KITTI, Latency benchmarking, Latency benchmarking, mAP, model-size benchmarking, Model-size benchmarking, PASCAL VOC, Precision/Recall, Synthetic data generation